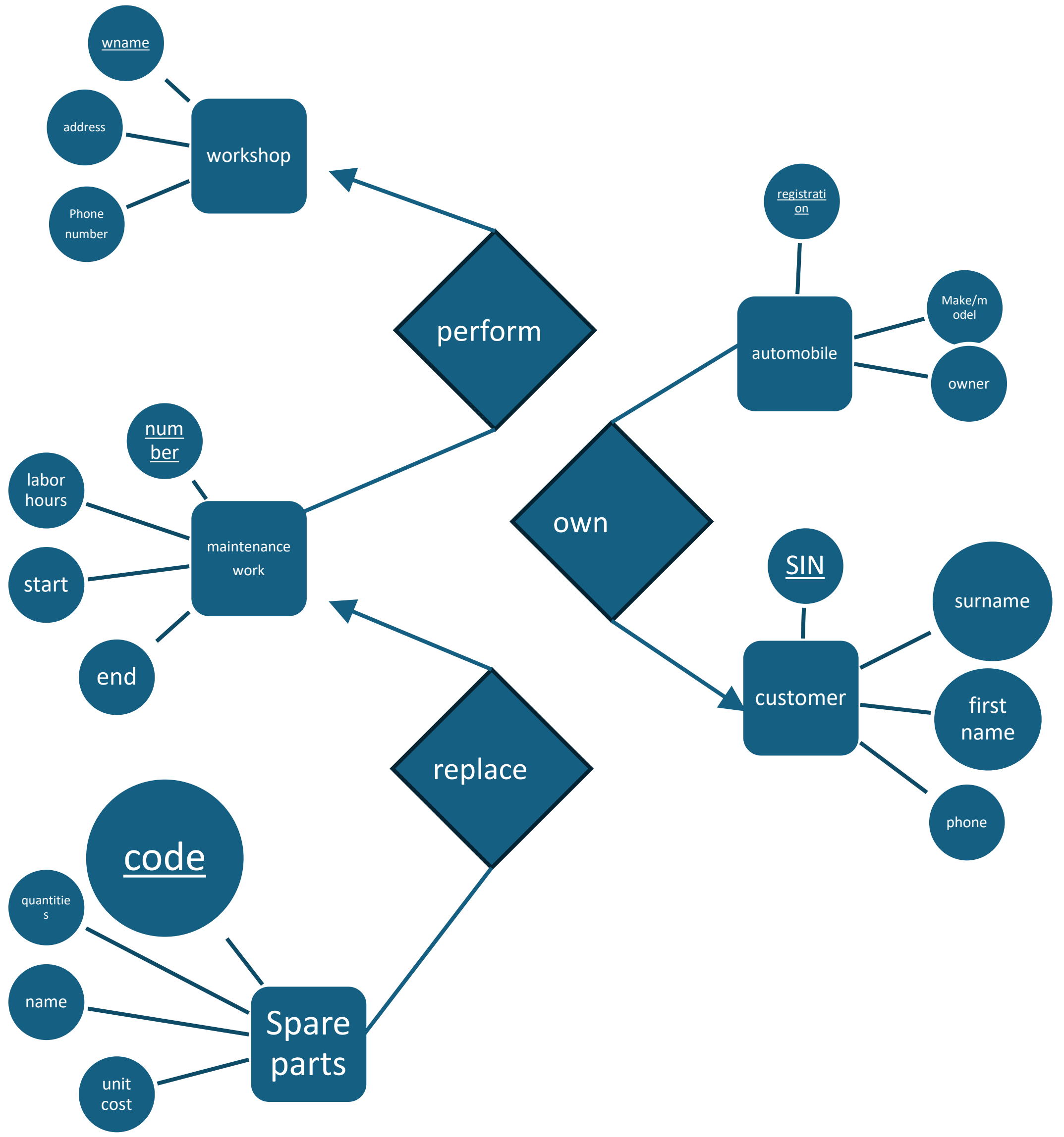
Q1



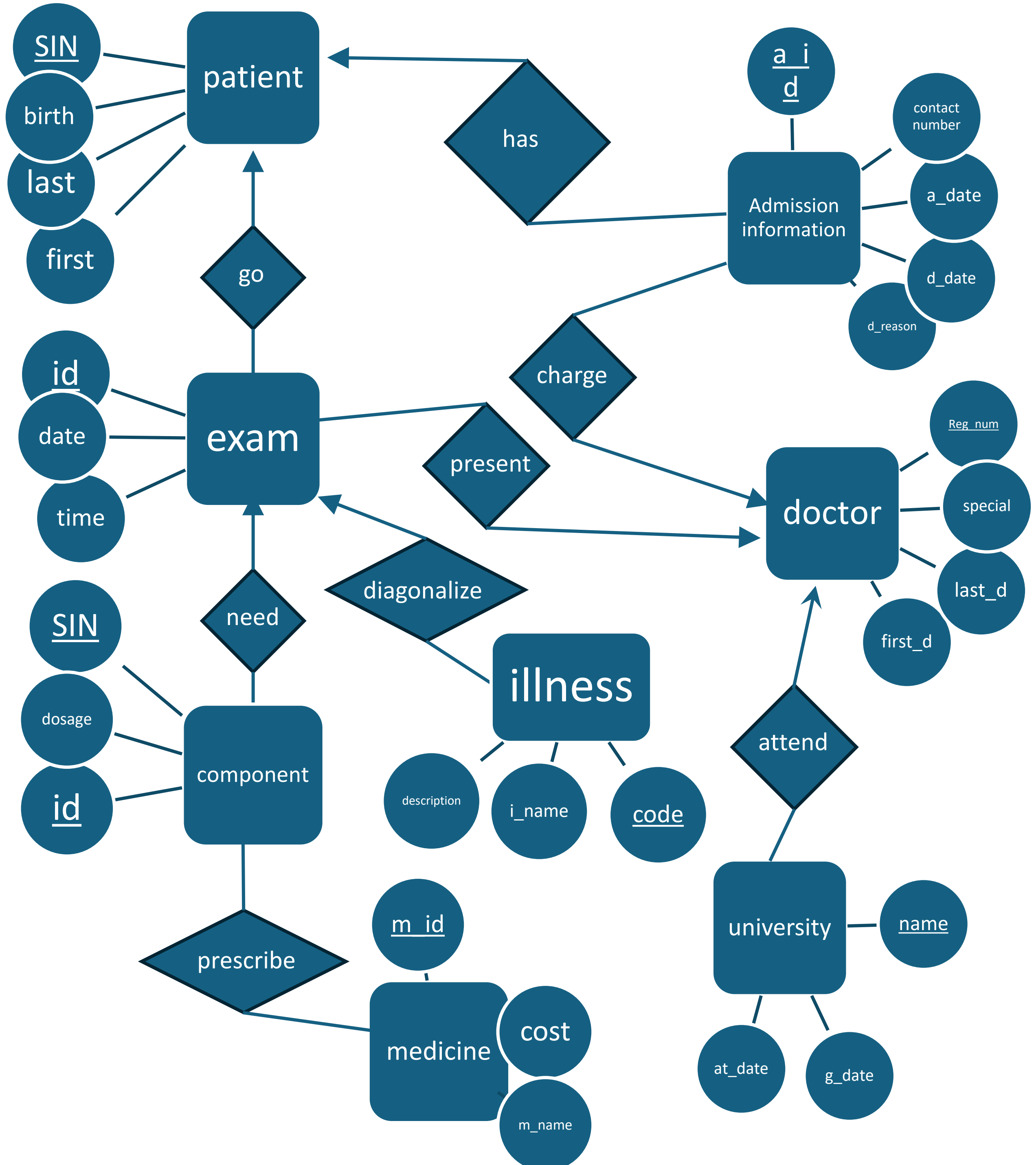
Relationship:

1. customer – own – automobile:
   1. One – to – Many
2. workshop – perform – maintenance work
   1. One – to - Many
3. maintenance work – replace – spare parts
   1. One – to – Many

Schema:

1. Workshop (‘<u>wname</u>’, ‘address’, ‘phone number’)
2. Maintenance work (‘<u>number’,</u> ‘start’, ‘end’, ‘labor hours’)
3. Spare parts (‘<u>code</u>’, ‘quantities’, ‘name’, ‘unit cost’)
4. Automobile (‘<u>registration</u>’, ‘make/model’, ‘owner’)
5. Customer (’<u>SIN</u>’, ‘surname’, ‘first name’, ‘phone’)

6. Perform (‘<u>wname</u>’, ‘number’)
7. Replace (‘<u>number</u>’, ‘code’)
8. Own (’<u>SIN</u>’, ‘registration’)



Q2

Relationship:

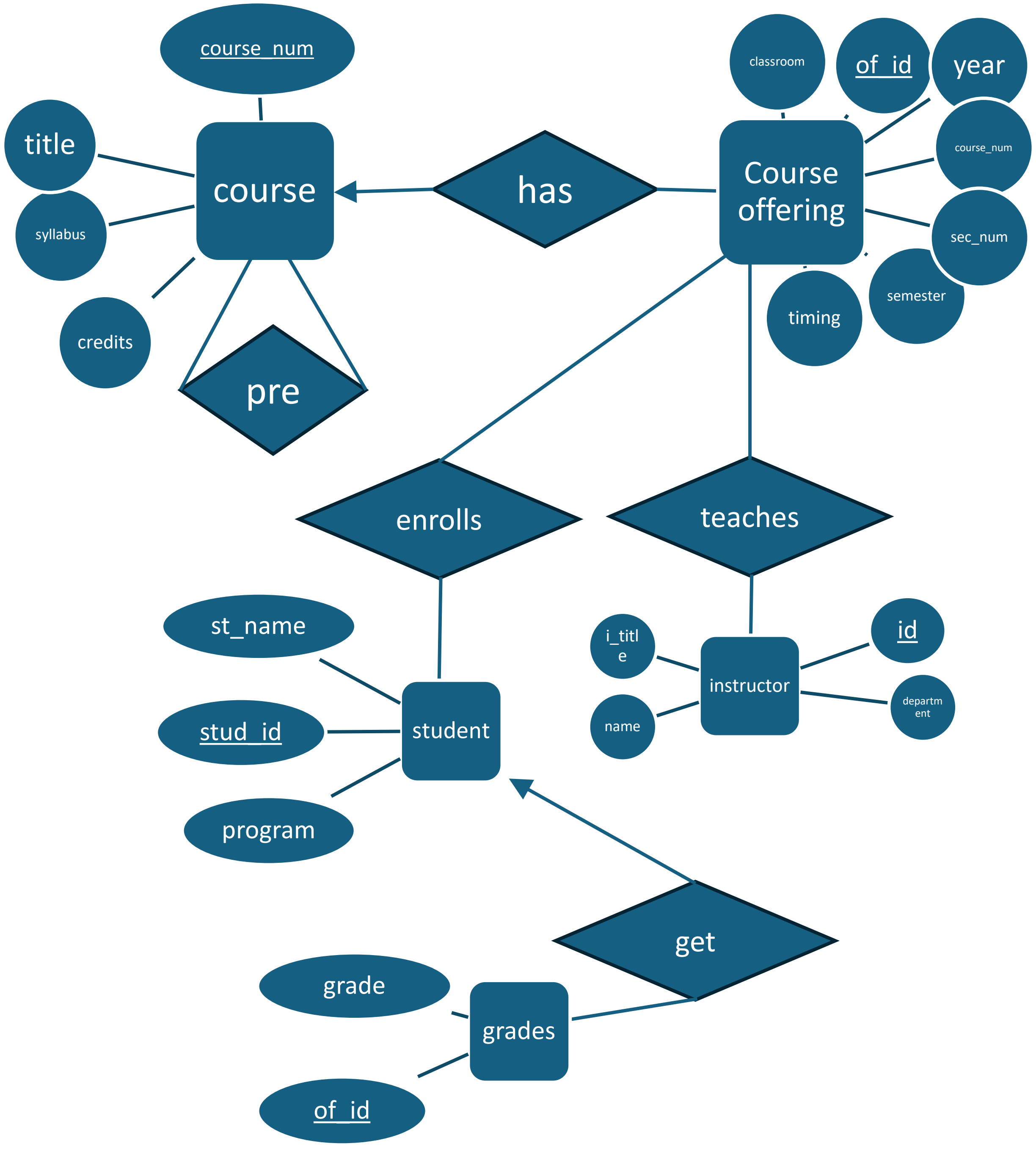
1. Patient – has – Admission information:
   1. One – to – Many
2. Doctor – charge - Admission information :
   1. One – to - Many
3. Doctor – attend – University:
   1. One – to – Many
4. Doctor – present – exam:
   1. One – to – Many
5. Patient – go – exam:
   1. One – to – Many
6. Exam – need – Component:
   1. One – to – Many
7. Component – prescribe – Medicine:
   1. Many – to - Many
8. Exam – diagonalize – illness:
   1. One – to - Many

Q2

Schema:

1. Patient (’SIN’, ‘first’, ‘last’, ‘birth’)
2. Exam (‘id’, ‘date’, ‘time’)
3. Component (’SIN’, ‘id’, ‘dosage’)
4. Medicine (’m_id’, ‘cost’, ‘m_name’)
5. Admission information (‘a_id‘, ‘contact number’, ‘a_date’, ‘d_date’, ‘d_reason’)
6. Doctor (‘reg_number’, ‘special’, ‘last_d’, ‘first_d’)
7. University (‘name’, ‘at_date’, ‘g_date’)
8. Illness (‘code’, ‘i_name’, ‘description’)
9. Has (‘SIN’, ‘a_id’)
10. Go (‘SIN’, ‘id’)
11. Need (‘SIN’, ‘id’, ‘dosage’)
12. Prescribe (‘id’, ‘dosage’, ‘m_id’)
13. Diagonalize (‘id’, ‘code’)
14. Present (‘reg_num’, ‘id’)
15. Charge (‘reg_num‘, ‘a_id‘)
16. Attend (‘reg_num’, ‘name’)

Q3



Relationship:

1. Course – has – Course offering:
   1. One – to – Many
2. Course – pre - Course:
   1. Many – to - Many
3. Course offering – teaches - instructor:
   1. Many – to – Many
4. Course offering – enrolls – student:
   1. One – to – Many
5. student – get – grades:
   1. One – to – Many

Q3

Schema:

1. Course (‘course_num’, ‘title’, ‘syllabus’, ‘credits’)
2. Pre (‘course_num1’, ‘course_num2’)
3. Has (‘course_num’, ‘of_id’)
4. Course offering (‘of_id’, ‘year’, ‘classroom’, ‘course_num’, ‘sec_num’, ‘semester’, ‘timing’)
5. Teaches (‘of_id’, ‘id’)
6. Instructor (‘id’, ‘department’, ‘i_title’, ‘name’)
7. Enrolls (‘of_id’, ‘stud_id’)
8. student (‘stud_id’, ‘st_name’, ‘program’)
9. Get (‘stud_id’, ‘of_id’, ‘grade’)
10. Grades (‘of_id’, ‘grade’)

Q4

1):

A -> B => A -> AB (Augmentation)
A -> C => AB -> BC (Augmentation)
A -> BC (Transitivity)

2):

BC -> B (Reflexivity)
BC -> C (Reflexivity)
A -> BC (Given)

A -> B (Transitivity)
A -> C (Transitivity)

3):

A -> B => AC -> BC (Augmentation)
BC -> D (Given)
AC -> D (Transitivity)

4):

1. A -> B (Given) => A -> AB (Augmentation)
2. AB -> CD (Given) & A -> AB
3. A -> CD (Transitivity)
4. A -> B (Given) & B ->DE (Given)
5. A -> DE (Transitivity)
6. C -> F (Given) => CD -> DF (Augmentation)
7. E -> G (Given) => DE -> DG (Augmentation)
8. A -> DF (Transitivity) & A -> DG (Transitivity)
9. DF -> F & DG -> G (Reflexivity)
10. A -> F & A -> G (Transitivity)
11. A -> AF & AF -> FG (Augmentation)
12. A -> FG (Transitivity)

Q5

1)

1. A -> B & B -> C => A -> C (Transitivity)
2. A -> B & B -> D => A -> D (Transitivity)
3. **Nontrivial FD’s**: A->B, B->C, B->D

4. A can determined all Attributes => **key is A**

5. **Super key:** AB, AC, AD, ABC, ABD, ACD, ABCD

2)

1. AB ->C => AB -> BC (Augmentation) & BC ->D => AB->D (Transitivity)
2. BC ->D => BC -> CD (Augmentation) & CD ->A => BC->A (Transitivity)
3. CD->A => CD->AD (Augmentation) & AD ->B = CD->B(Transitivity)
4. AD->B => AD -> AB (Augmentation) & AB ->C = AD -> C (Transitivity)
5. **Nontrivial FD’s**: AB->C, BC->D, CD->A, AD->B

6. From the above we get key: **AB, BC, CD, AD**

7. **Super Key**: ABC, ABD, BCD, ACD, ABCD

Q6

1)
1. A->BC => A->B & A->C (Decomposition)
2. A->B & B->D => A->D (Transitivity)
3. B->D => BC->CD (Augmentation) & CD-> E =>BC->E & A->BC =>A->E(Transitivity)
4. A+ = ABCDE => A is a key
5. For E->A => according to transitivity, E is also a key; For BC->E => according to transitivity, BC is also a key
6. **Keys :** E & A & BC & CD
7. **Reason:** There is no candidate key with three attributes b/c E or A or BC or CD enough to determinate all attributes. If adding any attribute, it is not the minimal length.

2)
1. B->D : B is not the super key. => **not BCNF**

3)
1. B->D: B is not the super key and D is a prime attribute. => **3NF**

Q6

4)
1. Attr(R1) ∪ Attr(R2) = Attr(R)
2. Attr(R1) ∩ Attr(R2) = A which is a key of R1 => **lossless**
3. R1: A->BC & R2: E->A, A->E, A->D
4. Can't preserve B->D => **not preserve**

5. Attr(R3) ∪ Attr(R4) = Attr(R)
6. Attr(R3) ∩ Attr(R4) = CD which is a key for R4 => **lossless**
7. R3: A->BC, A->D, B->D, BC->A, CD->A & R4: CD->E, E->C, E->D
8. Can preserve E->A, A->BC, CD->E, B->D => **preserve**

5)
R1: A->BC, BC->A: BC and A are a super key=> **Both**

R2: E->A, A->D, A->E both A & E are candidate keys and D is a prime attribute => **both**

R3: A->BC, A->D, B->D, BC->A, CD->A : B is not a super key and D, BC, A is a prime attribute => **3NF**

R4: CD->E, E->C, E->D: both CD and E are super key => **Both**

Q6

6)

1. For B->D, violate BCNF
2. B->D: R={ABCE, BD}

3. R1(ABCE), R2(BD)

4. preserve

Q7

1)
1. IS

2)
1. I->B: R={ISQD, IBO}
2. S->D: R={ISQ, IBO, SD}
3. R1(ISQ), R2(IBO), R3(SD)

4. Change to one attribute RHS {I->B, IS->Q, B->O, S->D}
5. Eliminate unnecessary attributes from LHS {I->B, IS->Q, B->O, S->D}
6. After operation the answer is {I->B, IS->Q, B->O, S->D}
7. I is not the super key => **not BCNF**

Q8

1)

1. $\pi_{\text{name}}(\sigma_{\text{sno=5}}(\text{students}))$
2. $\pi_{\text{address}}(\sigma_{\text{name='jones'}}(\text{students}) \bowtie \text{study} \bowtie \text{universities})$
3. $\pi_{\text{uname}}(\text{study} \bowtie \text{participate})$
4. $\pi_{\text{uname}}(\text{universities}) - \pi_{\text{uname}}(\text{study} \bowtie \text{participate})$
5. $\pi_{\text{eno}}(\text{took} - place \bowtie \text{participate} \bowtie \text{study})$

Q9

1)

1. $S := \sigma_{floor=2}(dept)$

   $\pi_{iname}(item) - \pi_{iname}(sales \bowtie_{sales.dept=s.dname} s)$

2. $M := \sigma_{eno=mgr}(employee)$ $E := \sigma_{eno \neq mgr}(employee)$. $S := \pi_{dept}(M \bowtie_{(M.dept = E.dept) \wedge (E.salary \geq M.salary)} E)$

   $\pi_{dept}(employee) - S$

3. $D := \sigma_{floor=2}(dept)$ $S1 := D \bowtie_{D.dname=sales.dept}(sales)$

   $\pi_{item}(\sigma_{count(dept) \geq 2}\, \gamma_{item,\ count(dept)}(S1))$

4. $S1 := \pi_{dept,\ item}(supply)$ $S2 := \pi_{dept,\ item}(sales)$ $S3 = S1 - S2$

   $\pi_{item}(S3 \bowtie_{S3.dept=S1.dept} S1)$

Q10

1)

1. S:= $\pi_{committee}(\sigma_{country='York'}(Representative))$ D:= $\pi_{president}(S \bowtie Committies)$

   $\pi_{firstname,\ surname}(D \bowtie_{D.president=Representative.number} Representative)$

2. S:= $\sigma_{name='Finance'}(Committies)$

   $\pi_{firstname,\ surname}(S \bowtie_{S.number=Representative.committee} Representative)$

3. S:= $\sigma_{name='Finance'}(Committies)$. D:=$S \bowtie_{S.number=Representative.committee} Representative$ K:=$D \bowtie_{D.county=Counties.code} Counties$

   $\pi_{firstname,\ surname,\ county,\ region}(K)$

4. S:= $Representative \bowtie_{Representative.country = Countries.code} Countries$

   S1 := S

   $\pi_{region}(\sigma_{S.surname=S1.surname}) _{\wedge S.number \neq S1.number}(S \times S1)$